Security Door lock System

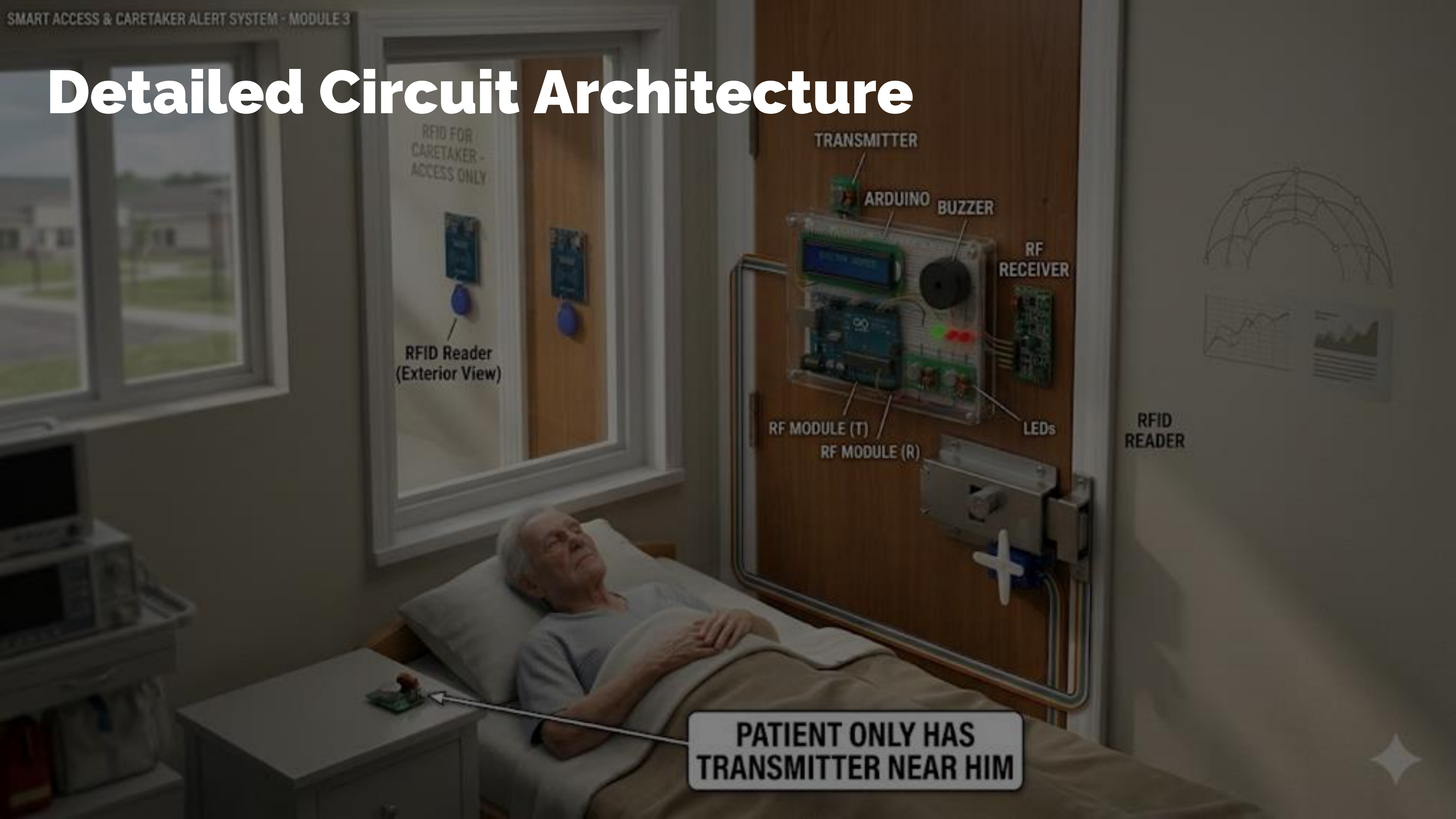
Problem Statement & Objective

The Challenge: For bedridden, elderly, or physically impaired individuals, getting out of bed to physically unlock a door for family members or medical staff causes immense strain.
My Solution: A budget friendly prototype built on the **Arduino UNO** that lets a patient safely open their door directly from their pillow using a wireless remote, while allowing caregivers local tap access via RFID.

Core Hardware Architecture

- Arduino UNO R3
- 433MHz RF Transmitter & Receiver pair (Bypasses walls and obstacles).
- MFRC522 RFID Kit (Proximity badge scanning at the door).
- 16x2 LCD Module with I2C backpack.
- Red & Green LEDs paired with a 5V Active Buzzer for instant sensory feedback.

Detailed Circuit Architecture



Component Category	Component	Pin Label	Connect To (Arduino/Breadboard)	Notes
Logic & Data	**I2C LCD 16x2**	**GND**	Breadboard GND (-)	Common Ground
		VCC	Breadboard 5V (+)	
		SDA	**Analog Pin A4**	I2C Data Line
		SCL	**Analog Pin A5**	I2C Clock Line
Authentication	**MFRC522 RFID Reader**	**VCC**	**Arduino 3.3V Pin**	**Warning: 3.3V ONLY!**
		RST	**Digital Pin D9**	Reset
		GND	Breadboard GND (-)	Common Ground
		MISO	**Digital Pin D12**	SPI Pin
		MOSI	**Digital Pin D11**	SPI Pin
		SCK	**Digital Pin D13**	SPI Pin
		SDA (SS)	**Digital Pin D10**	Slave Select
Long Range	**433MHz RF Receiver**	**VCC**	Breadboard 5V (+)	
		DATA	**Digital Pin D2**	**MUST be D2 (Interrupt 0)**
		GND	Breadboard GND (-)	Common Ground
Indicators	**Green LED (D3)**	Long Leg	**Digital Pin D3**	
	Red LED (D4)	Long Leg	**Digital Pin D4**	
	Active Buzzer (5V)	Positive Leg	**Digital Pin D5**	Short leg to GND
Mechanical Actuation	**SG90 Micro Servo Motor**	**Brown Wire**	Breadboard GND (-)	Ground
		Red Wire	**Arduino 5V Pin**	Power
		Orange Wire	**Digital Pin D6**	Servo PWM Signal

System Execution Logic

Patient-Initiated Access: The patient uses a handheld **433MHz RF remote** from their bedside. Because the system utilizes radio frequency (RF) signals, the command penetrates walls and obstacles, allowing the user to unlock the door without physical effort.
Caregiver Access: Visiting medical staff or family use an **RFID key fob**, which they simply tap against an external reader to gain entry.

The Arduino validates both inputs against pre stored secure data:
Access Granted: The system confirms a match, updates the **LCD display**, illuminates the **Green LED**, sounds a confirming **double beep**, and triggers a **servo operated latch** to unlock the door for 5 seconds before automatically relocking.
Access Denied: If an unauthorized token is detected, the system immediately rejects entry by flashing the **Red LED**, sounding a **continuous alarm buzzer**, and displaying an "Invalid Token" error on the LCD.

Future Scope & Mechanical Completion

- High-Torque Actuation (MG996R)
- **Deadbolt Latch System**
- Transitioning to an ESP32 architecture to enable real-time entry timestamp

Thanks for watching

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